GUILLERMO SANCHEZ

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Portfolio & technical blog: guillesanper.github.io

EDUCATION

Universidad Complutense de Madrid Madrid, Spain
B.Sc. in Software Engineering Sep 2022 – Expected Jun 2027

- **Bachelor's thesis (in progress, team of two):** *Music Similarity Analysis and Graph Generation using Siamese Neural Networks and Alternative Classification Techniques*. Siamese networks (contrastive and triplet loss) learn song embeddings from log-mel spectrograms, MERT embeddings and acoustic descriptors, against classical classifiers, MFCC and MERT. Similarity graphs evaluated on FMA and MagnaTagATune (Recall@K, MAP, nDCG) with paired bootstrap and Holm correction. Python, PyTorch.
- Electives focused on machine learning, evolutionary computation, knowledge engineering and big data.

IES San Isidoro de Sevilla
Bachillerato (Spanish high school diploma), Sciences 2020 – 2022

PROJECTS

Sentinel — Intrusion detection platform *Python, XGBoost, Terraform, Kubernetes, GCP*

- Built a two-stage detector on CICIDS2017: an Isolation Forest fitted on benign traffic filters flows, and XGBoost classifies what survives.
- Deployed on GKE with Terraform modules (VPC, IAM, cluster), kustomize overlays and default-deny network policies; training runs as Kubernetes Jobs, inference behind FastAPI.
- Moving to a chronological split and running control experiments exposed errors in my own published results (one F1-macro figure off by ~4×, a safety-critical threshold metric off by 14×); published each correction.

Zero-knowledge Password Manager — Self-hosted *Django, TypeScript, PostgreSQL, Redis, MinIO, Docker*

- Password and file manager whose server cannot read what it stores: the master password is turned into keys in the browser (Argon2id + HKDF).
- The server holds only a verifier and AES-256-GCM blobs bound to their database row; rotating the password rewraps a 32-byte key instead of re-encrypting the vault.
- Rebuilt the original design after threat-modelling a compromised database, moving key derivation out of the server.

DataMap — Database migration platform *FastAPI, TypeScript, PostgreSQL, Redis, Docker*

- Introspects PostgreSQL, MySQL, SQL Server and Oracle schemas through their system catalogs, maps tables and columns in a graph UI and generates INSERT INTO … SELECT scripts.
- Orders migrations with a topological sort over the foreign-key graph that fails explicitly on cycles.
- Schema mapping never reads user rows; execution is opt-in, backs up target tables first and supports a dry run that rolls back. Stored credentials are encrypted with a derived, never-stored key.

ReactorGuard — Anomaly detection for a simulated nuclear reactor *Python, Kafka, Kubernetes, Terraform*

- Energy-conserving point-kinetics model of a pressurised water reactor that generates labelled process and instrument faults.
- Sensor-fault detectors run on a Kafka stream; GKE and Kafka defined with Terraform and Strimzi. ML alerting layer in progress.

SKILLS

Languages: Spanish (native); English (B2 First, Cambridge, 2020).

Programming: Python, TypeScript, Java, C++, SQL.

ML & data: NumPy, pandas, scikit-learn, XGBoost, PyTorch.

Backend & web: FastAPI, Django, Node.js/Express, Spring, React.

Infrastructure: Linux, Docker, Kubernetes, Terraform, Kafka, Google Cloud (GKE), CI pipelines, PostgreSQL, Redis.